[B Tech CSE Core Sec-A]

ASSIGNMENT-2

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(DBMS) [Section - A]

① Fill in the Blanks:-

Q1. The Entity-Relationship (ER) model visually represents how entities relate in a database, offering a clear blueprint of the data's organizational structure.

Q2. In the Relational model, data is organized into tables, with relationships between entities defined through foreign key constraints.

Q3. The Object oriented model treats data as a collection of objects, each with unique attributes and methods, mirroring real world entities and interactions.

Q4. Integrity constraints ensure the reliability of data by enforcing rules that must be adhered to, maintaining the accuracy of information.

[SECTION-B]

Case study based :-

② In a healthcare system, discuss how data abstraction can simplify the interaction between healthcare professionals and patient records.

Ans. Data abstraction in DBMS hides the complex details of how data is stored while providing users with only the information they need.

→ Physical level (Internal)

- Hides how patient records are physically stored.
- eg. Patient records may be stored in encryped files across servers, using indexes for fast access.
- Healthcare staff don't need to know storage formats or encryption details; they just query data.

→ Logical level (Conceptual)

- Show what data exists and how it's related.

- eg Relationships like :-
  Patient → Doctor
  Patient → Appointment
  Patient → Prescription
- Professionals can understand the overall structure without worrying about storage details.

→ View level (External)

- Provides customized views for different roles.
- eg # Doctor sees patient history, allergies, current medications.

Nurse sees vitals and medication schedule

Billing staff insurance details and charges.

- Simplifies interaction and prevents access to unnecessary/sensitive information.

③ Explore the potential benefits and challenges associated with implementing data abstraction in this content.

Ans. Benefits :-

1) Simplified Access :- Staff access only the data relevant to their role.

2) Enhanced security :- Sensitive patient data can be hidden from unauthorized roles.

3) Data independence :- changes in physical storage or database structure do not affect user interfaces.

4) Improved productivity :- reduces time spent navigating complex data structure.

Challenges :-

1) Implementation complexity :- Designing multiple levels of abstraction requires careful planning.

2) Performance overhead :- Views and abstraction layers may slightly reduce query speed.

3) Training requirement :- staff may need training to use views efficiently

4) Integration issues :- integrating multiple healthcare systems with different database schemas can be difficult.

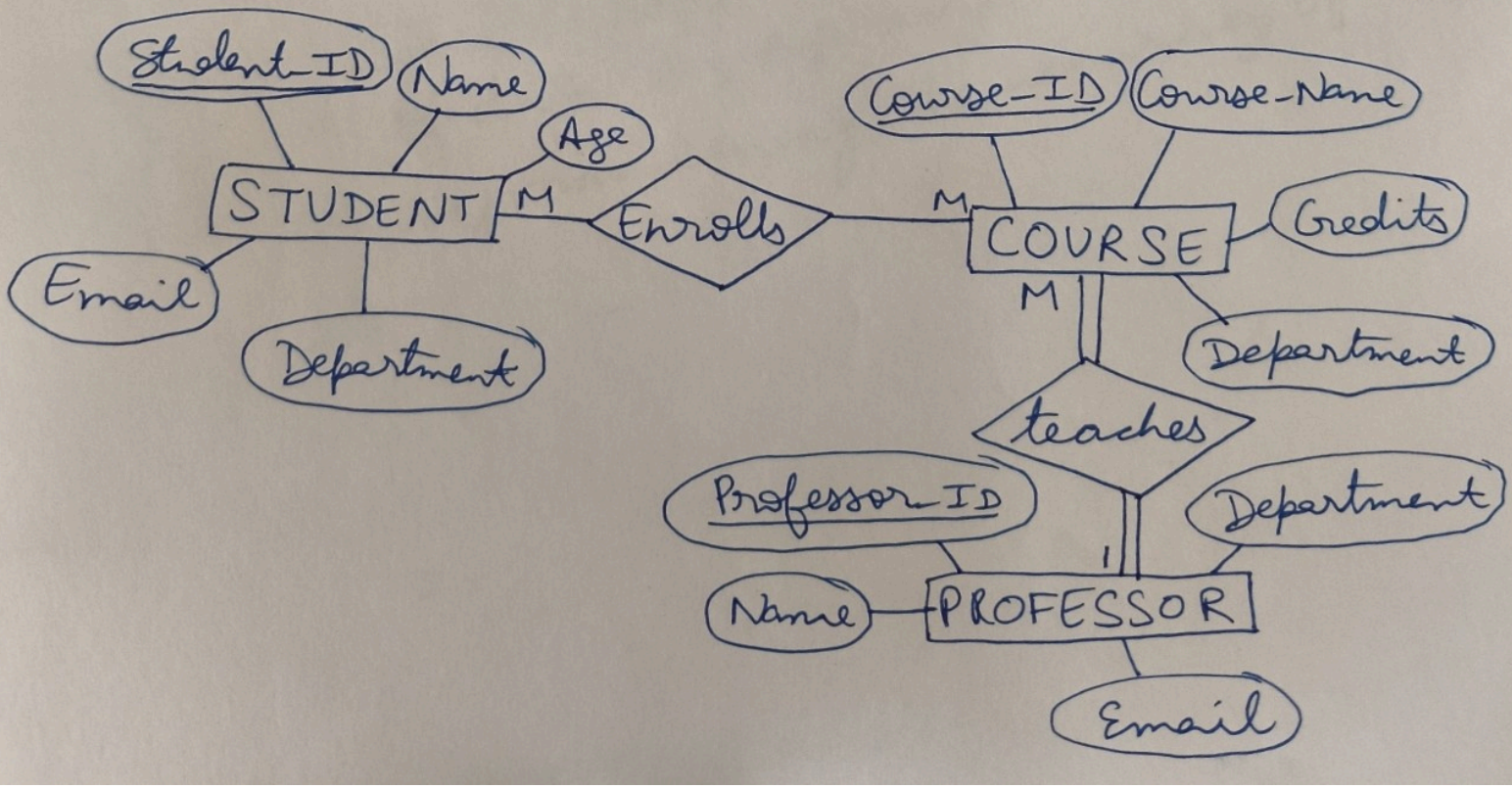
5) Maintenance :- changes in requirements may require updates to views and abstractions

[SECTION-C]

(4) Design a database for a university, incorporating entities like students, courses, and professors. Using the entity-relationship between these entities. Include attributes for each entity to capture relevant information such as student names, course details and professor information.

Ans.

Entities	Attributes
Student	Student_ID, Name, Age, Email, Department
Course	Course_ID, Course_Name, Credits, Department
Professor	Professor_ID, Name, Department, Email.



Relationship	Cardinality	Participation
Enrolls	Many-to-many	Student - Course
Teaches	One-to-many	Professor - Course